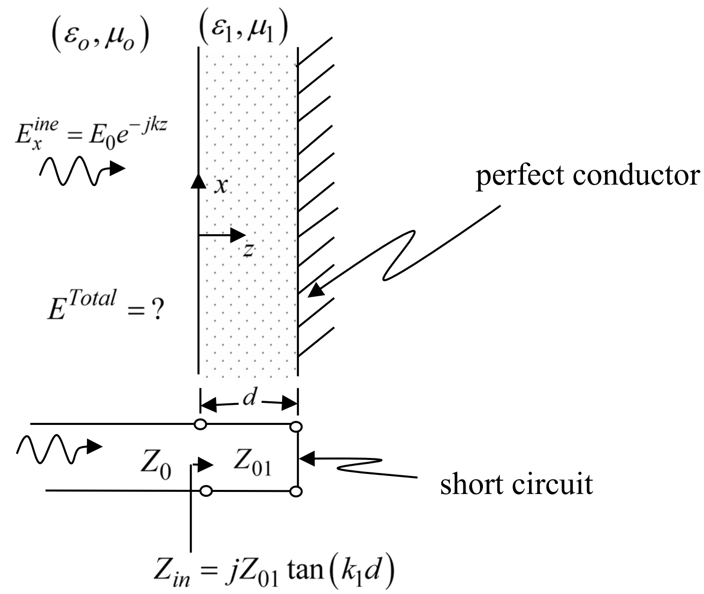
ECEN 635 Homework #5

We are covering topics similar to those in Chapter 5 in Balanis but adding the very useful transmission line technique.

1. As a review of the transmission line mathematics redo the example at the end of notes 4 appendix. Change the line length to 7 meters and answer parts (a) through (d). Do not skip the step that requires analytically finding power delivered to the load.

2.(a) First find an expression for the transmission line voltage in the short circuited transmission section $0 \le z \le d$ in the figure below. (b)Then use duality to find an expression for the electric field in the dielectric slab region $0 \le z \le d$. Define all variables in your answer in terms of given known qualities (e.g. $E_0$). Use the coordinate system defined in the figure above.




3. Using transmission line methods for the specific case of an impedance matching region shown below: (a) First use the coordinate system shown in the figure below and create the transmission line dual for the plane wave normally incident on a dielectric shown below (b) next solve the TL problem for the voltage and current in regions 2 and 3. (c) then using transmission line/plane wave duality give your answer in terms of the electromagnetic fields in regions 2 and 3. (d) Find expressions for the time average power in regions 1 and 3 and compare.